Bangladesh University of Business and Technology (BUBT)

Department of Computer Science and Engineering

Midterm Examination: Summer 2026

Course Code: CSE 341 | Course Title: Advanced Programming Language

Intake: 54 | Program: B.Sc. Engg. in CSE

Time: 1.5 Hours | Marks: 30

Answer all the questions

CO1: Implement advanced programming concepts to identify and resolve errors and use structured knowledge representations for effective problem-solving.

[CO1, PO1] 1. (a). Identify and correct all errors with debugging techniques in the given Java program to achieve successful program execution. **[5]**

```
class Vehicle {

    final void startEngine() {
        System.out.println("Engine Started");
    }
}

class ElectricCar extends Vehicle {

    void startEngine() {
        System.out.println("Electric Engine Started");
    }
}
```

```
public class SmartCity {

    public static void main(String[] args) {

        final int speedLimit = 80;
        speedLimit = 100;

        int population = "50000";

        double temperature = 35.5;
        int temp = temperature;

        int[] zones = new int[4];
        System.out.println(zones[6]);

        Building b = new Building();
    }
}
```

Figure-1(a): Java Code

(b) A mobile recharge system allows customers to recharge in different ways. A customer may recharge by entering only the amount or by entering both the mobile number and the amount. Implement static polymorphism in Java to handle these different recharge options. Utilize “this” keyword where necessary and display the recharge details. **[5]**

CO1: Implement advanced programming concepts to identify and resolve errors, and use structured knowledge representations such as diagrams for effective problem-solving.

[CO1, PO1] 2. (a). A programmer wants to represent different cuboid-shaped objects in a Java program. Initiate a program using inheritance where a superclass named Rectangle contains the properties length and breadth, a constructor to initialize them, and a method to calculate the area. Then create a subclass named Cuboid that inherits from Rectangle and adds an extra property named height. The Cuboid class should have a constructor that initializes length, breadth, and height using constructor chaining, and a method to calculate the volume. In the main method, create an array of three Cuboid objects, initialize them with different values, and display the length, breadth, height, area, and volume of each cuboid. **[5]**

(b) PayEase is a digital payments app that allows users to pay using different payment methods. Every payment method performs one common action named processPayment(amount), but the calculation varies depending on the payment type. MobileBanking adds a 1% service charge to the given amount, CreditCard adds a flat 20 transaction fee, and Cash charges exactly the given amount without any extra fee. Create a Java program using a superclass named PaymentMethod and three subclasses named MobileBanking, CreditCard, and Cash. Each subclass must override the processPayment(amount) method according to its own payment rule. In the Main class, apply dynamic method dispatch to achieve late binding and display the final payable amount for each payment method. **[5]**

CO2: Analyze relationships and architectural decisions in real-world problems using OOP concepts such as inheritance, polymorphism, abstraction, and interfaces.

[CO2, PO2] 3. (a). Examine the provided diagram to interpret its structure and relationships and implement a corresponding Java program. **[5]**

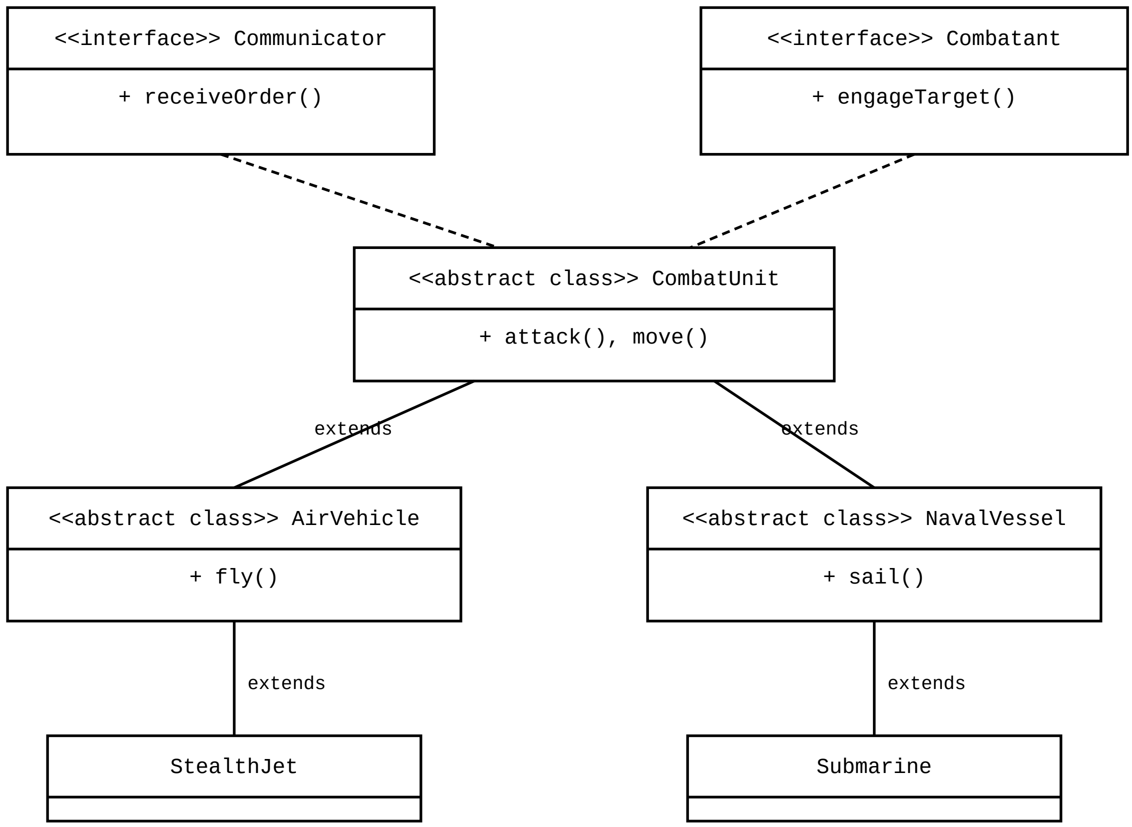

Figure-2(a): UML Class Diagram

(b) Evaluate the design trade-offs in a secure banking system by reviewing a refactoring of SavingsAccount (base class) and PremiumAccount (subclass). Assess the use of strict data hiding by making the balance field private, initialized via constructor, and accessed only through getBalance(), and critique the security implications of including or omitting a setter method in preserving financial invariants. Further interpret the class hierarchy and evaluate how PremiumAccount extends functionality with bonus interest, ensuring proper initialization using the “super” keyword. **[5]**